Zijin Zhang

zijin@ucla.edu • beforerr.github.io/beforerr • 0000-0002-9968-067X

Education

Ph.D. **University of California, Los Angeles (UCLA)**, Ph.D. in Planetary Science 2022 – 2026
- Thesis: Kinetic-scale solar wind current sheets: statistical characteristics and their role in energetic particle transport
- Advisors: Prof. *Vassilis Angelopoulos* and Prof. *Anton Artemyev*

B.Sc. **University of Science and Technology of China (USTC)**, B.Sc. in Space Physics 2018 – 2022
- Thesis: Kinetic simulation of the interaction between the Moon's magnetic anomalies and the solar wind (DOI: 10.13140/RG.2.2.15841.68968)
- Advisor: Prof. *Xin Tao*

University Corporation for Atmospheric Research, NASA's Living with a Star Heliophysics Summer School 2024

Research Interests

- Magnetosphere-ionosphere coupling: energetic particle precipitation and wave-particle interactions
- Heliophysics: Solar wind current sheets and energetic particle transport
- Computational plasma physics and space physics environment data analysis

Publications

The JuliaSpacePhysics ecosystem: High-performance, modular, and scalable software for space physics In preparation
Zhang, Z., Angelopoulos, V, et al.

Energetic particle transport driven by solar wind current sheets In preparation
Zhang, Z., Artemyev, A. V., Angelopoulos, V.

Separating Field-Line Curvature and Wave Scattering in Nightside Energetic Electron Precipitation With ELFIN In preparation
Zhang, Z., Shen, Y., Artemyev, A. V., Angelopoulos, V.

High-Frequency EMIC Wave Power Inferred From Relativistic Electron Precipitation: An ELFIN–PWING Conjunction Survey Submitted
Zhang, Z., Shiokawa, K., Artemyev, A. V., Mourenas, D., Angelopoulos, V.

Properties of current sheets in two-dimensional tearing-mediated magnetohydrodynamic turbulence 2026
Shi, C., Velli, M., Sioulas, N., **Zhang, Z.**
10.3847/1538-4357/ae742a (Astrophysical Journal)

Peculiarities of precipitating electron spectra revealed through a combined DMSP-ELFIN data set 2026
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Zhang, X.-J.
10.1029/2026JA035299 (Journal of Geophysical Research - Space Physics)

Multifluid equilibrium model of current sheets with interpenetrating ion beams 2026
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Velli, M.
10.1063/5.0331297 (Physics of Plasmas)

Comparison of solar wind current sheets in the inner heliosphere 2026
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Shi, C., Phan, T.
10.3847/1538-4357/ae46a1 (Astrophysical Journal)

Quantification of ion scattering by solar-wind current sheets: Pitch-angle diffusion rates 2025
Zhang, Z., Artemyev, A. V., Angelopoulos, V.
10.1103/pkzv-k5t3 (Physical Review E)

Solar wind discontinuities in the outer heliosphere: Spatial distribution between 1 and 5 AU 2025
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Vasko, I.
10.1029/2025JA034039 (Journal of Geophysical Research - Space Physics)

Relativistic electron flux decay and recovery: Relative roles of EMIC waves, chorus waves, and electron injections 2024
Zhang, Z., Artemyev, A., Mourenas, D., Angelopoulos, V., Zhang, X.-J., et al.
10.1029/2024JA033174 (Journal of Geophysical Research - Space Physics)

A search for technosignatures around 11680 stars with the Green Bank Telescope at 1.15–1.73 GHz 2023
Margot, J.-L., Li, M. G., Pinchuk, P., … **Zhang, Z.**
10.3847/1538-3881/acfda4 (Astronomical Journal)

Awards

- Betty and S.L. Huang graduate fellowship (2026)
- Outstanding student scholarship, USTC (2022)
- Outstanding student scholarship, USTC (2021)
- Social practice (field investigation in natural sciences) individual & team awards (2021)
- Outstanding student scholarship, USTC (2019)

Software

SPEDAS.jl: Julia-based space physics environment data analysis software (DOI: 10.5281/zenodo.15181866)

PlasmaBO.jl: Efficient plasma electromagnetic dispersion-relation solver (DOI: 10.5281/zenodo.18058843)

PlanetaryMagneticFields.jl: A unified framework for planetary magnetic field modeling (DOI: 10.5281/zenodo.18428922)

Other Experience

Artificial Intelligence of Things Lab, Undergraduate Research Assistant 2021 – 2022
- Implemented a distributed system to monitor edge devices and automate IT deployment and management

University of California, Los Angeles, Teaching Assistant
- EPS SCI M7: Perils of Space — Introduction to Space Weather (2025)
- EPS SCI 9: Solar System and Planets (2025)

University of Science and Technology of China
- Teaching Assistant, Optics and Thermodynamics (2022)
- Student President, USTC Student Scientific Expedition Association (2021-2022)